

Statistics II

Multiple Linear Regression Model

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- General idea of multiple linear regression (MLR)

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- Interpretation of coefficients
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- R^2 vs. adjusted R^2
- Global F test and individual t tests
- A practical example with two explanatory variables

Population Model (with k regressors)

Specification

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + \varepsilon_i, \quad i = 1, \dots, n$$

- Y_i : dependent variable (response)

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Linear Relationship Assumption

The marginal effect of X_j on $E[Y|X]$ is constant and equal to β_j , *holding the other regressors fixed*.

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Y : price; X_1 : size (m^2); X_2 : age (years).

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β_1 : how the average price changes with one additional m^2 , *for properties of the same age*.

β_2 : how the average price changes with one additional year of age, *for properties of the same size*.

Coefficient Interpretation: Real Estate Example

Model

$$\text{price} = \beta_0 + \beta_1 \cdot \text{size} + \beta_2 \cdot \text{age} + \varepsilon$$

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Estimation via OLS

Minimization Problem

$$\min_{\beta_0, \dots, \beta_k} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^k \beta_j x_{ji} \right)^2$$

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- Error variance estimate: $\hat{\sigma}^2 = \frac{SSE}{n - k - 1}$

R^2 and Adjusted R^2

R^2

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

- Measures the fraction of variability in Y explained by the model.
- **Never decreases** when regressors are added.

Adjusted R^2

$$\bar{R}^2 = 1 - \frac{SSE/(n - k - 1)}{SST/(n - 1)}$$

- Penalizes models with many variables.
- Useful for comparing models with different numbers of regressors.

ANOVA and Global F Test

Source	SS	df	MS	Stat.
Regression	SSR	k	$MS_{Reg} = SSR/k$	$F = \frac{MS_{Reg}}{MS_{Error}}$
Error	SSE	$n - k - 1$	$MS_{Error} = SSE/(n - k - 1)$	
Total	SST	$n - 1$		

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$H_0 : \beta_1 = \dots = \beta_k = 0$ vs. H_a : at least one $\beta_j \neq 0$.

- Reject H_0 if $F > F_{k, n-k-1; \alpha}$ (or if the p-value $< \alpha$).

Individual t Tests and Confidence Intervals

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- Test statistic: $t = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \sim t_{n-k-1}$ under H_0
- $(1 - \alpha)$ CI for β_j :

$$\hat{\beta}_j \pm t_{n-k-1; 1-\alpha/2} \text{SE}(\hat{\beta}_j)$$

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Remark

Avoid **strong multicollinearity** among regressors; it inflates standard errors and hampers identification of partial effects.

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Example: Responses to an Advertisement

responses	circulation (thousands)	ad size
12	150	10
6	130	5
8	110	10
4	100	5
9	115	10
13	160	5
23	170	5
18	140	5
15	130	10
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7	110	10
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Typical Output (Excel/R)

Multiple $R \approx 0.812$

$R^2 \approx 0.659$; $\bar{R}^2 \approx 0.591$

Standard error ≈ 3.37

F test: p-value ≈ 0.00

Circulation: $p \approx 0.001$

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- Conclusion: keep *circulation*; reconsider *ad size* (not significant).

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- Compare models using adjusted R^2 and the F test.

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- Always check residual diagnostics before drawing conclusions.
- Compare models using adjusted R^2 and the F test.
- Next lecture: variable selection and regularization (concepts).

Thank you! Questions?